

1- Context and Motivation

- As MT systems performance has greatly improved, fine-grained metrics to evaluate specific aspects of translation are needed
- Terminology is an important and critical aspect in translation, including MT (e.g. MQM error typology)
- Terminology-specific metrics that take into account necessary criteria, i.e., accuracy, consistency, and domain conventions, are needed

Type	Example
Src (En)	On Wednesday, the World Health Organization (WHO) declared the ongoing outbreak of COVID-19 — the disease caused by coronavirus SARS-CoV-2 — to be a pandemic .
Tgt (Fr)	Mercredi, l'Organisation mondiale de la santé (OMS) a qualifié l' épidémie en cours de COVID-19 (la maladie provoquée par le coronavirus SARS-CoV-2) de pandémie .
Transf	Mercredi, l'Organisation mondiale de la Santé (OMS) a déclaré que l' éclosion actuelle de COVID-19 de la maladie causée par le coronavirus CoV-2 SRAS) était une pandémie .
Src (En)	By converting the phase detected by such a method to angle of projected sheet beam , there can be obtained three-dimensional shapes of the object in real time .
Tgt (Ja)	こうして、検出された位相を投影シートビームの角度に変換することにより、対象物の 3次元形状 が実時間で得られる。
Gemma2	その方法で検出された 相 を投影された シートビーム の 角度 に変換することで、 リアルタイム で 対象物の3次元形状 を得ることができる。

Table 1: Examples in English-French and English-Japanese. Type: Src and Tgt respectively represent the source and its target reference. MT: Transformers (Transf) and Gemma2. Correct translations are shown in green, acceptable translations in blue, and wrong translations in red

3- Experiment: Dataset

- English-French:** Tico-19 test set used in WMT 2021
 - 30 documents translated from English into 36 languages
 - 2,100 parallel sentences with terminology annotations
- English-Japanese:** ASPEC-JE parallel corpus (Nakazawa et al, 2016)
 - includes abstracts of 24 scientific domains
 - 1,300 sentences with term annotations

Test set	# Terms	# Distinct
Tico-19 (En)	2,557	215
Tico-19 (Fr)	5,879	324
ASPEC (En)	4,288	2,397
ASPEC (Ja)	5,656	2,899

Table 2: Statistics of Tico-19 and ASPEC tests

4- Experiment: Results

Best Scores:	Related work	Positive CoTERM	Negative CoTERM	Integrated CoTERM									
En → Fr													
	NMT						LLM						
	Transf	Conv	LightC	DynC	Opus	NLLB 3B	T5-L 770M	MBart 610M	M2M 1B	Madlad 7B	Llama4 109B	Gemma3 27B	Qwen2 7B
BLEU	42.94	28.23	36.94	29.18	44.00	48.20	41.71	40.66	42.21	45.52	47.45	47.43	36.36
chrF	68.06	62.53	65.94	64.43	69.04	72.22	66.32	67.05	62.24	70.96	71.91	71.92	64.94
BertScore	90.79	90.66	90.68	90.84	91.13	91.89	90.69	90.56	90.67	91.23	91.77	91.60	89.44
COMET	86.06	85.22	85.66	85.95	85.98	87.28	85.88	85.45	85.93	87.56	87.46	87.89	83.60
EM	85.60	83.94	81.83	81.11	91.64	96.00	85.68	92.25	94.69	95.54	96.31	95.79	89.26
WO2	20.36	20.16	20.05	20.38	20.18	25.31	20.22	20.90	22.71	23.38	24.79	24.27	20.37
F1	72.70	74.26	71.99	71.45	76.26	87.01	70.62	76.65	84.20	81.54	87.28	88.85	79.97
TC	65.52	56.06	63.12	61.59	65.97	68.42	60.85	58.03	61.74	70.42	72.55	71.01	55.76
CoTERM _f ⁺	67.02	56.25	62.79	61.39	67.79	74.08	61.21	60.36	64.51	74.44	75.93	72.20	55.11
CoTERM _f ⁺	56.68	45.70	52.10	51.62	56.15	62.06	52.15	50.66	53.75	62.26	64.04	61.07	47.42
CoTERM _f ⁺ _d	77.04	66.97	73.18	71.93	80.32	83.80	74.85	77.61	81.50	84.12	84.82	84.48	73.77
CoTERM _f ⁺ _d	68.06	58.51	63.70	65.72	69.87	75.44	65.18	68.55	73.61	74.69	76.21	76.79	66.77
CoTERM ⁻	30.62	31.50	34.26	33.07	43.85	41.26	34.49	25.46	35.07	41.58	42.66	46.45	34.81
CoTERM _f	48.82	43.87	48.52	47.23	55.82	57.67	47.85	42.91	49.79	58.01	59.29	59.32	44.96
CoTERM _r	43.65	38.60	43.18	42.34	50.00	51.66	43.32	38.06	44.41	51.92	53.35	53.76	41.11
CoTERM _f _d	53.83	49.23	53.72	52.50	62.08	62.53	54.67	51.53	58.28	62.85	63.74	65.46	54.29
CoTERM _r _d	49.34	45.00	48.98	49.39	56.86	58.35	49.83	47.00	54.34	58.13	59.43	61.62	50.79

Table 3: Results of the evaluation of NMTs and LLMs on the Tico-19 test set for English-French

En → Ja													
NMT						LLM							
	Fugumt	Tako	Tatoeba	W-X	NLLB 3B	M2M 12B	mBART 610M	Gemma2 27B	Gemma3 27B	Llama3 70B	Llama4 109B	Qwen2 7B	
BLEU	27.53	21.02	11.05	27.30	17.84	14.37	19.21	23.86	23.92	20.98	24.97	23.39	
chrF	37.15	31.22	21.56	37.97	27.66	26.38	29.99	34.96	35.34	32.91	32.26	38.24	
BertScore	86.25	84.57	77.87	87.15	82.33	78.86	83.97	85.81	86.21	85.53	86.48	85.98	
COMET	88.80	86.30	80.11	90.27	85.60	83.11	86.90	89.82	90.22	89.18	89.77	86.99	
EM	92.23	91.10	80.04	93.83	88.51	89.72	91.27	92.12	92.49	89.78	92.36	86.23	
F1	55.00	51.18	38.50	56.62	47.62	46.81	50.49	55.11	55.63	52.71	55.76	50.80	
TC	40.10	35.53	25.87	41.24	32.31	32.13	35.28	39.98	40.26	38.38	40.50	38.03	
CoTERM _f ⁺	41.28	36.80	26.66	42.17	33.14	33.02	37.07	40.56	40.62	38.39	41.11	36.00	
CoTERM _f ⁺	38.83	33.99	23.99	39.78	30.89	30.69	34.21	38.35	38.70	36.26	38.96	35.02	
CoTERM ⁻	42.17	48.09	40.42	45.95	46.54	50.01	45.60	44.30	43.42	42.53	44.42	38.67	
CoTERM _f	-0.44	-5.64	-6.88	-1.89	-6.70	-8.49	-4.26	-1.87	-1.40	-2.07	-1.65	-1.33	
CoTERM _r	-1.67	-7.05	-8.21	-3.08	-7.82	-9.66	-5.69	-2.97	-2.36	-3.13	-2.73	-1.82	

Table 4: Results of the evaluation of NMTs and LLMs on the ASPEC test set for English-Japanese

2- Proposed Metric

- CoTERM is based on the Herfindahl-Hirshman Index (HHI) and extends the Consistency index (C_t) from Itagaki et al (2007)

$$HHI = \sum_{i=1}^n (MS_i)^2 \quad C_t = \frac{1}{p} \sum_{j=1}^p \sum_{i=1}^n \left(\frac{f_i}{k_j} \times 100 \right)^2$$

MS: Market Share
n: number of companies

p: number of products
n: number of translations of t
f_i: frequency of translation i
k_j: total translation occurrences of t

- Takes into account: **Accuracy**, **Consistency** at the segment level and **Appropriateness** to the domain conventions (rigid vs flexible CoTERM)
- Operates at the **segment level** (paragraph, document, corpus...)

Positive CoTERM

$$CoTERM^+ = \frac{1}{S} \sum_{s=1}^S \left[\frac{1}{T_s} \sum_{t=1}^{T_s} \sum_{i=1}^{n_s} \left(\frac{f_i^t}{|t|} \times 100 \right)^2 \right]$$

→ Incorporates accuracy to the consistency index C_t

S1.	The spread of covid-19 keeps increasing.	S: number of segments
R1.	La <u>propagation</u> du <u>covid-19</u> ne cesse d'augmenter.	T _s : number of source terms t to translate
C1.	La <u>propagation</u> du <u>covid-19</u> continue d'augmenter.	n _s : number of translations
S2.	The WHO supports the spread of clean technologies .	t : number of occurrences of the source term t
R2.	L' <u>OMS</u> soutient la <u>diffusion</u> de technologies propres.	
C2.	L' <u>OMS</u> encouragent la <u>diffusion</u> de technologies propres.	
S3.	The WHO has declared that the ongoing outbreak of covid-19 is a pandemic with high spread .	
R3.	L' <u>OMS</u> a déclaré que l'épidémie en cours du <u>covid-19</u> est une pandémie avec un risque élevé de propagation rapide.	
C3.	L' <u>OMS</u> a déclaré que l' éclosion en cours du <u>covid-19</u> était une pandémie avec une forte dissémination.	

Negative CoTERM

$$CoTERM^- = \frac{1}{T} \sum_{t=1}^T \sum_{i=1}^n \left(\frac{f_i^t}{|t|} \times 100 \right)^2$$

→ Observes whether the **erroneous terms** are generated consistently or randomly

T: number of source terms t to translate
n: number of wrong translations

S.	The WHO has declared that the ongoing outbreak of covid-19 is a pandemic .
R.	[L' <u>OMS</u> a déclaré que] l'épidémie [en cours du <u>covid-19</u>] est une pandémie.
C.	[L' <u>OMS</u> a déclaré que] l' éclosion [en-cours du <u>covid-19</u>] était une pandémie.

Integrated CoTERM

$$CoTERM = \begin{cases} \frac{1}{2}(a + b) & \text{if } a > b \\ \frac{1}{2}(a - b) & \text{if } a \leq b \end{cases}$$

→ Combines positive (a) and negative (b) CoTERM

5- Inter Annotator Agreement

	Kappa (κ)			Pearson		
	A12	A13	A23	A12	A13	A23
Term level						
Flexible	0.50	0.69	0.50	0.81	0.91	0.82
Rigid	0.54	0.73	0.52	0.82	0.93	0.83
Sent level	0.21	0.18	0.14	0.62	0.63	0.57

Table 5: Inter-Annotator Agreement. A12: agreement between the raters A1 and A2

- High IAAs for all the pairs
- The overall rating is reliable
- IAAs for term level rating is higher than for the sentence level rating

6- Correlations with Human Judgments

- Correlation between human raters and CoTERM at the term level
- For each set of sentences in which a term occurs, we compute CoTERM⁺ and the mean average score of the raters
- The ASPEC sample contains 309 En-Ja term pairs
- Moderate to strong correlations between human raters and CoTERM

Metric	Kendall			Pearson		
	A1	A2	A3	A1	A2	A3
CoTERM _r ^f	0.43	0.41	0.43	0.49	0.46	0.48
CoTERM _r ^f	0.47	0.45	0.46	0.57	0.51	0.55

Table 6: Correlations with human judgments on ASPEC En-Ja

7- Conclusion

- A new set of measures and a framework to evaluate MT systems with regard to their ability to properly translate terms
- Empirically observed several pretrained NMT and LLM models with respect to their ability in translating terms
- Better scores in generic metrics such as BLEU or BertScore do not necessarily mean better performance in term translations
- Importance of using fit-for-purpose evaluation metrics
- Rigid and flexible CoTERM provide a means to compare MT models with respect to term translations